

Validation of Satellite-Derived Daily Latent Heat Flux over the South China Sea, Compared with Observations and Five Products

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ABSTRACT

This study describes the development of the South China Sea (SCS) daily satellite-derived latent heat flux (SCSSLH) for the period of 1998–2011 at $0.25^\circ \times 0.25^\circ$ resolution using data mainly from the Tropical Rain Measuring Mission (TRMM) Microwave Imager (TMI). Flux-related variables of daily TMI data smoothed with 3-day running mean were finally chosen because of the best fit with the 1727 high-quality observations from seven moored stations and 24 ship surveys. Near-surface air specific humidity was computed using the global relationship based on satellite precipitable water. Verification against 1016 high-resolution radiosonde profiles from 1998 to 2012 and the time series from the Xisha automatic weather station during 2008–10 indicate that this satellite-derived air specific humidity can reasonably capture observed mean condition and temporal variability. They are therefore used to derive SCSSLH based on the Coupled Ocean–Atmosphere Response Experiment version 3.0 (COARE 3.0) algorithm. Compared with five other latent heat flux products—the Goddard Satellite-Based Surface Turbulent Fluxes version 2 (GSSTF2), the objectively analyzed air–sea heat fluxes (OAFlux), the Hamburg Ocean Atmosphere Parameters and Fluxes from Satellite Data version 3 (HOAPS3), the National Centers for Environmental Prediction/Department of Energy Global Reanalysis 2 (NCEP-2), and the European Centre for Medium-Range Weather Forecasts (ECMWF)—the daily SCSSLH shows the highest spatial resolution and realistic values in the SCS, with an exception along the northern continental shelf. More importantly, the other five products seem to overestimate the latent heat flux systematically. The flux representation in this study comes not only with a better flux algorithm but also with the improved estimates of bulk variables based on in situ measurements, which further highlights the unique role of high-quality meteorological measurements and atmospheric weather stations in evaluating the air–sea interaction in the SCS.

1. Introduction

The heat fluxes at the air–sea interface include shortwave, longwave, latent heat, and sensible heat fluxes. Among the four components, shortwave and latent heat

fluxes are the dominant ones over the tropical oceans, where the latent heat flux is comparable to the shortwave flux.

An important application of passive microwave data from satellites is to improve latent heat flux estimates over data-sparse areas of the global oceans. Remote sensing, however, cannot directly measure latent heat flux, so an indirect approach through a bulk formula has been proposed using satellite retrievals of wind speed and humidity differences between sea skin and 10-m height (Schlüssel 1995; Kubota et al. 2002; Sun

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et al. 2003; Yu and Weller 2007, Liu and Curry 2006; Liu et al. 2011a,b).

The estimate of latent heat flux is obtained through the following two steps: first, to obtain the estimates of flux-related surface meteorological variables; and second, to choose a bulk flux algorithm selected to compute the latent heat flux. The accuracy of these meteorological parameters, including sea surface temperature (SST) T_s , wind speed U , and surface air humidity q_a , is critical for the estimation of latent heat flux. Among these input parameters, q_a is the most important and difficult one. Liu and Niiler (1984) demonstrated that over periods longer than a few weeks, mean precipitable water W is an adequate predictor of q_a . Liu (1986) proposed a polynomial relationship to compute q_a as a function of W (q – W relationship), which was refined later using a more extended dataset of radiosonde measurements. Hsu and Blanchard (1989) further verified the q – W relationship using radiosonde data from field experiments conducted in regions of different geography. Liu (1988) evaluated the statistical relationship using ship reports and an atmospheric general circulation model. Jourdan and Gautier (1995) compared global latent heat flux computed from satellite-derived W , air temperature, and Liu's q – W relationship. Liu et al. (1999) noted that the observed radiances at the frequencies of the Microwave Imager on board the satellite for the Tropical Rain Measuring Mission (TRMM; TMI) are sensitive to the change of latent heat flux at all seven channels. Their results also demonstrated the importance of the TMI's low-frequency channels, which are absent in the Special Sensor Microwave Imager (SSM/I) data.

In previous studies, the estimations and evaluations were carried out for the global oceans. The critical parameter W of all available satellite-derived latent heat flux products comes from SSM/I, including products developed by the objectively analyzed air–sea fluxes (OAFlux) project, the Goddard Satellite-Based Surface Turbulent Fluxes version 2 (GSSTF2), the Japanese Ocean Flux Data Sets with Use of Remote Sensing Observations (J-OFURO), and the Hamburg Ocean Atmosphere Parameters and Fluxes from Satellite Data version 3 (HOAPS3). Tomita and Kubota (2006) analyzed the accuracy of J-OFURO latent heat flux through comparisons with moored buoy data. They found that estimation errors were mainly caused by the accuracy of q_a . Jiang et al. (2005) pointed out that latent heat flux from OAFlux was more accurate than those from the National Centers for Environmental Prediction–National Center for Atmospheric Research Global Reanalysis (NCEP-1) and the NCEP/Department of Energy Global Reanalysis 2 (NCEP-2) in the tropical Atlantic, especially outside of the equatorial region.

There is no latent heat flux product produced specifically for the South China Sea (SCS), which is an important part of the East Asian monsoon system. Because of the lack of measurements, it is also difficult to produce accurate latent heat flux product. Liu (1992) pointed out that the q – W relationship does not work very well at synoptic and shorter time scales, since the vertical distribution of water vapor in the atmosphere is only coherent for a period longer than a week. None of the stations selected by Liu (1986) is located in the SCS, so uncertainty in this area remains. In this paper, we will evaluate Liu's q – W relationship in the SCS using high-resolution radiosonde data collected in 1997, 1998, and from 2004 to 2012, as well as other input parameters from TMI instead of SSM/I. As good validation standard is sparse in the SCS, our continuous measurements in this region should make a significant contribution, not only in the present study but in future validation. The measurements and methods used to estimate latent heat flux and related parameters are described in section 2, and evaluations are presented in section 3. Section 4 compares different daily latent heat flux products with observations. Section 5 summarizes our main findings and discusses their implications.

2. Data and algorithm

a. Retrieval algorithm

The bulk expression of the latent heat flux is

$$Q_{LH} = \rho_a L_e C_e U (q_s - q_a), \quad (1)$$

where ρ_a is the air density; L_e is the latent heat of evaporation; C_e is the turbulent exchange coefficient for latent heat, which is a function of stability and height; U is the surface wind speed relative to the sea surface; and q_s and q_a are the sea surface and near-surface atmospheric specific humidity, respectively. Note that q_s is computed from the saturation humidity q_{sat} for pure water at T_s :

$$q_s = 0.98 q_{sat}, \quad (2)$$

where the factor of 0.98 is used to take into account the reduction in vapor pressure caused by nonpure water with a typical salinity of 34 psu (Fairall et al. 1996a,b; Zeng et al. 1998).

The procedure was applied to each of the three independent variables (i.e., U , T_s , and q_a) from which the latent heat flux was computed using the new algorithm, the Coupled Ocean–Atmosphere Response Experiment version 3 (COARE 3.0, available online at http://www.coaps.fsu.edu/COARE/flux_algor/), developed from the

TABLE 1. List of selected meteorological observations used in the validation.

Station or cruise		Lat (°N)	Lon (°E)	No.	Deployment period
Mooring buoys and stations	SCSMEX_buoy SCS1	8.10	115.60	25	17 Apr–13 Jun 1997 and 10 Apr–4 May 1998
	SCSMEX_buoy SCS3	12.98	114.41	199	8 Jun–23 Dec 1998
	SCSMEX_buoy SCS3'	13	114	255	13 Apr–23 Dec 1998
	SCSMEX_mooring KX	6.41	110.26	33	7 May–20 Jun 1998
	SCSMEX_mooring SY	20.60	117.40	38	6 May–23 Jun 1998
Automatic station	Xisha automatic station	16.83	112.33	138	6 Apr 2008–31 Aug 2010
SCSIO cruises	SCSMEX_R/V <i>Kexue1</i> *	1.16–36.03	103.53–122.29	58	18 Apr–30 Jun 1998
	SCSMEX_R/V <i>Shiyan3</i> *	18.45–21.57	112.26–118.32	68	25 Apr–21 Jul 1998
	SCSMEX_R/V <i>Haijian74</i>	6.75–20	108–119	55	22 Apr–5 May 1998
	SCSMEX_R/V <i>Xyh14</i>	6.75–20	108–119	53	12–27 Jun 1998
	Open cruise 2004	14.39–22.19	109.48–119.45	18	17 Sep–4 Oct 2004
	Open cruise 2005	13.38–22.24	109.48–119.61	18	5–22 Sep 2005
	Open cruise 2006*	17.98–22.25	109.47–120.01	20	9–28 Sep 2006
	Open cruise 2007*	17.95–23.39	110.06–120.03	27	13–27 Aug 2007
	Open cruise 2008*	17.01–22.41	109.50–119.98	64	16 Aug–4 Sep 2008
	Open cruise 2009*	15.77–22.57	109.55–120.09	57	1–19 Sep 2009
	Open cruise 2010*	16.93–23.05	110.30–120.01	64	2–20 Sep 2010
	Open cruise 2011*	16.97–22.32	109.44–120.02	37	21 Aug–10 Sep 2011
	Nansha cruise 2007*	5.38–20.87	108.49–117.98	40	15 May–20 2007
	Xisha cruise 2011*	16.47–18.39	109.56–111.74	39	2–5 Jun 2011
	MEL cruise 2006*	10.50–22.15	110.20–119.00	11	26 Nov–16 Dec 2006
	Cruise 200709*	18.01–25.01	111.13–119.75	21	24–29 Sep 2007
	Cruise 200803*	18.89–22.70	112.64–113.85	15	16–20 Mar 2008
	SCOPE-PP cruise 2008*	20.27–23.53	114.07–117.96	31	29 Jun–14 Jul 2008
	YZD cruise 2009*	18.21–33.72	109.49–122.55	36**	9 Jun–5 Jul 2009
	Rose cruise 2009*	9.19–19.55	110.31–117.96	41**	19–30 Jun 2009**
Other cruises	Rose cruise 2010*	7.06–19.00	109.95–117.95	38**	26 Oct–11 Nov 2010**
	IO cruise 2010*	0.01–21.83	79.61–113.82	63**	14 Apr–27 May 2010**
	IO cruise 2011*	0.01–20.20	80.01–113.73	47**	2 Apr–13 May 2011**
	IO cruise 2012*	5.94°S–19.72°N	79.77–113.51	26**	25 Feb–19 Apr 2012**

* Cruises with high-resolution upper-air sounding observations, while the others provide only surface meteorological measurements.

** Observation numbers and the time period operated in the SCS are statistical only.

Tropical Ocean and Global Atmosphere COARE (TOGA COARE; Fairall et al. 1996b, 2003). COARE 3.0 was based on published results and 2777 one-hour covariance flux measurements in the National Oceanic and Atmospheric Administration (NOAA) Environmental Technology Laboratory (ETL) inventory. Compared with COARE 2.6a, COARE 3.0's range of wind speed validity is extended to 0–20 m s⁻¹. The updates also include improving the stability function, shortening the stability iteration, and eliminating the need for a Webb correction to latent heat flux. Based on direct turbulent flux measurements determined from 12 ship cruises over the tropical and midlatitude oceans, Brunke et al. (2003) showed that the overall performance of the COARE 3.0 algorithm tops the other 11 algorithms considered.

In Eqs. (1) and (2), U , T_s , q_s , and q_a are independent variables and they are to be estimated from the synthesis. According to COARE 3.0, input data for the synthesis include not only U , T_s , q_s , and q_a but also other surface meteorological variables (air temperature, solar, and infrared radiative fluxes), which we take from NCEP-2. All

other variables, including L_e and C_e , are determined based on the information of the input variables and the specified parameterizations.

b. Remote sensing dataset

The q - W relationship of Liu (1986) is in the form of a polynomial based on the monthly summaries of atmospheric soundings taken at 49 stations on small islands and weather ships over the major oceans over 17 years. The fifth-order polynomial regression (a global relationship) is as follows:

$$q_a = 3.818\,724W + 0.189\,721\,9W^2 + 0.189\,189\,3W^3 - 0.075\,490\,36W^4 + 0.006\,088\,244W^5, \quad (3)$$

where W is the total precipitable water in centimeters and q_a is in grams per gram. Zeng et al. (2009) and Zeng and Wang (2009) have calculated monthly and weekly satellite-derived latent heat flux in the SCS based on COARE 3.0 algorithm and this q - W relationship. To

ensure the accuracy, verification of this relationship on daily time scale in the SCS will be processed prior to the development of daily latent heat flux.

Variables W , T_s , and U are taken from TMI, which is a five-spectral (10.65, 19.35, 21.3, 37.0, and 85.5 GHz), nine-channel passive microwave radiometer. Two kinds of daily TMI datasets are used: the daily mean TMI maps (separated into ascending and descending segments, called daily TMI data) and the smoothed daily mean TMI, after applying a 3-day running mean (datasets are averaged over 3 days and time is defined according to the last day of data contributing to the average, called smoothed daily TMI data). The three input parameters are T_s , U , and W . Two 10-m surface wind speed retrievals are available, derived primarily from the 11- and 37-GHz channels. We use the 11-GHz surface wind speed maps only, because they have less missing data than the 37-GHz maps.

c. Meteorological observations

Table 1 shows the observations used to examine Liu's q - W relationship and TMI input parameters, which include moored buoys, automatic weather station, and cruise observations (also see Fig. 1). These measurements were taken during the following field experiments: moored buoys and stations from the South China Sea Monsoon Experiment (SCSMEX); the Xisha automatic weather station, operated by the South China Sea Institute of Oceanology (SCSIO) since 2008; the mooring station buoy QF301, operated by the South China Sea Marine Prediction Center; the State Oceanic Administration; and ship observations obtained by 24 cruises, mainly operated by the SCSIO.

The SCSMEX buoy data are 15-min-averaged measurements. In an effort to evaluate the performance of the satellite products on time scales as short as daily, buoy bulk variables are averaged again over one day using those 15-min measurements. The mooring station KX from Research Vessel (R/V) *Kexue1* is located at 6.3°N, 110°E, and its 0.5-h-averaged observation period is from 7 May to 20 June 1998. The mooring station SY from R/V *Shiyan3* is located at 20.4°N, 116.9°E, operating from 6 May to 23 June 1998. The mooring station of buoy QF301 is settled along the northern continental shelf, providing daily mean records of surface wind speed, wind direction, air pressure, air temperature, relative humidity, SST, and wave height from March to May 2011 (92 days in all). The Xisha automatic weather station, operated by the SCSIO since 2008, is located at 112.33°N, 16.82°E. Every 2 min, the surface wind speed, wind directions, air temperature, relative humidity, and air pressure are automatically recorded by this weather station. The observational data from April 2008 to August 2010 are averaged again over each day for the

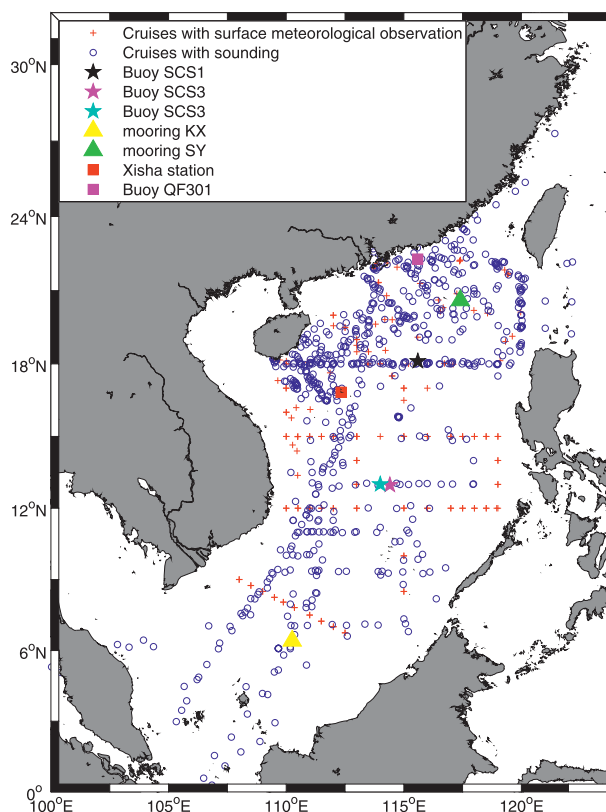


FIG. 1. Locations of selected meteorological and radiosonde observations used in this study. Ship observations are marked by solid squares and cruises with high-resolution upper-air sounding observations by open circles. Three buoys are marked by stars, while two moorings by triangles.

calculation. To facilitate the comparison, all ship observations are treated as moored data in each day and averaged the same way as the moored buoys; that is, the errors caused by the ship movement during each day are neglected in this study. More information on the observations used in this study is shown in Table 1.

Atmospheric soundings were taken during 20 cruises, which are marked by asterisks in Table 1. During all of these cruises, high-resolution soundings were launched from a height of 3–10 m above the sea surface. During the intensive observational period of the SCSMEX, there were 154 and 152 high-resolution soundings taken by R/V *Shiyan3* and R/V *Kexue1*, respectively. Data are obtained from Vaisala global positioning system (GPS) radiosondes using an H-Humicap humidity sensor. High-resolution balloon soundings were taken during the SCSIO open cruises in 2006–11, Nansha and Xisha cruises, State Key Laboratory of Marine Environmental Science (MEL) cruises, cruises 200709 and 200803, the Northern SCS Coastal Oceanographic Process Experiment Pilot Project (SCOPE-PP)

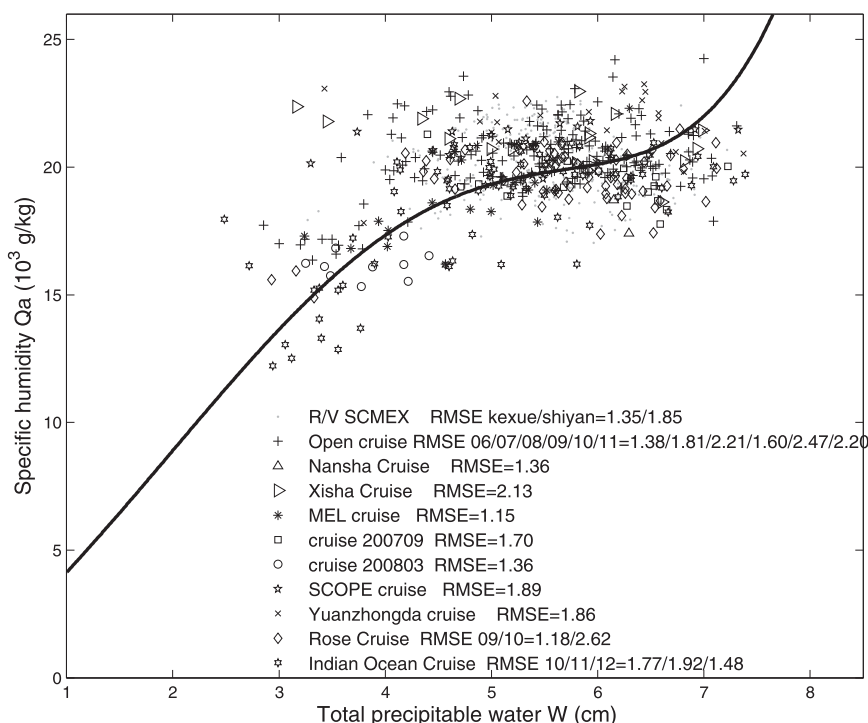


FIG. 2. Scatter diagram of q_a (g kg^{-1}) vs W (cm) obtained from 401 high-resolution sounding profiles. Solid curve is based on Eq. (3). The RMSE of each cruise is given in the bottom-right corner.

cruise, Yuanzhongda (YZD) cruise, Rose cruise 2009–10, and the Indian Ocean (IO) cruise 2010–12. These cruises contributed a total of 1016 sounding profiles for the validation of q – W parameterization in the SCS. We discard those profiles that did not reach 3000 m.

d. Other daily latent heat flux products

We also include four other daily latent heat products for comparison. The first one is the GSSTF2 (Byers et al. 2003), which has a spatial resolution of $1^\circ \times 1^\circ$. The GSSTF2 flux model is a bulk aerodynamics algorithm based on the Monin–Obukhov similarity theory, including salinity and cool-skin effects. The methods for deriving the GSSTF2 daily latent heat flux and input parameters essentially are the same as those of Chou et al. (1997). Daily turbulent fluxes are derived using the GSSTF2 bulk flux model from daily mean values of the SSM/I U of Wentz (1997) and SSM/I q_a , and T_s from NCEP-2. The second one is the OAFflux (Yu et al. 2004) with $1^\circ \times 1^\circ$ resolution, which was computed from the COARE algorithm 2.6a using basic surface meteorological variables, including U (Wentz 1997) and q_a (Chou et al. 1997) from the SSM/I, T_s from the Advanced Very High Resolution Radiometer (AVHRR),

and the surface meteorological variables from the European Centre for Medium-Range Weather Forecasts (ECMWF) operational forecast model and NCEP-2. The third one is HOAPS3 (Andersson et al. 2010), which has a spatial resolution of $0.5^\circ \times 0.5^\circ$. Except for the SST dataset, all variables are derived from SSM/I passive microwave satellite data using intersatellite calibration with homogeneous and reliable spatial and temporal fields. The other two are reanalysis products from the ECMWF and NCEP-2.

3. Validation of Liu's algorithm and TMI parameters

a. Relationship between W and q_a

A standard method based on Byers (1974) and List (1984) is employed to obtain an estimate of W from sounding measurements by integrating the humidity soundings with the trapezoidal rule up to the highest level reported. At every level, W is a product of layer-averaged vapor density (obtained using dewpoint temperature) times the layer thickness.

Figure 2 shows q_a versus W obtained from 1016 sounding profiles during the cruises mentioned above. The solid curve is based on Eq. (3). To delineate the differences

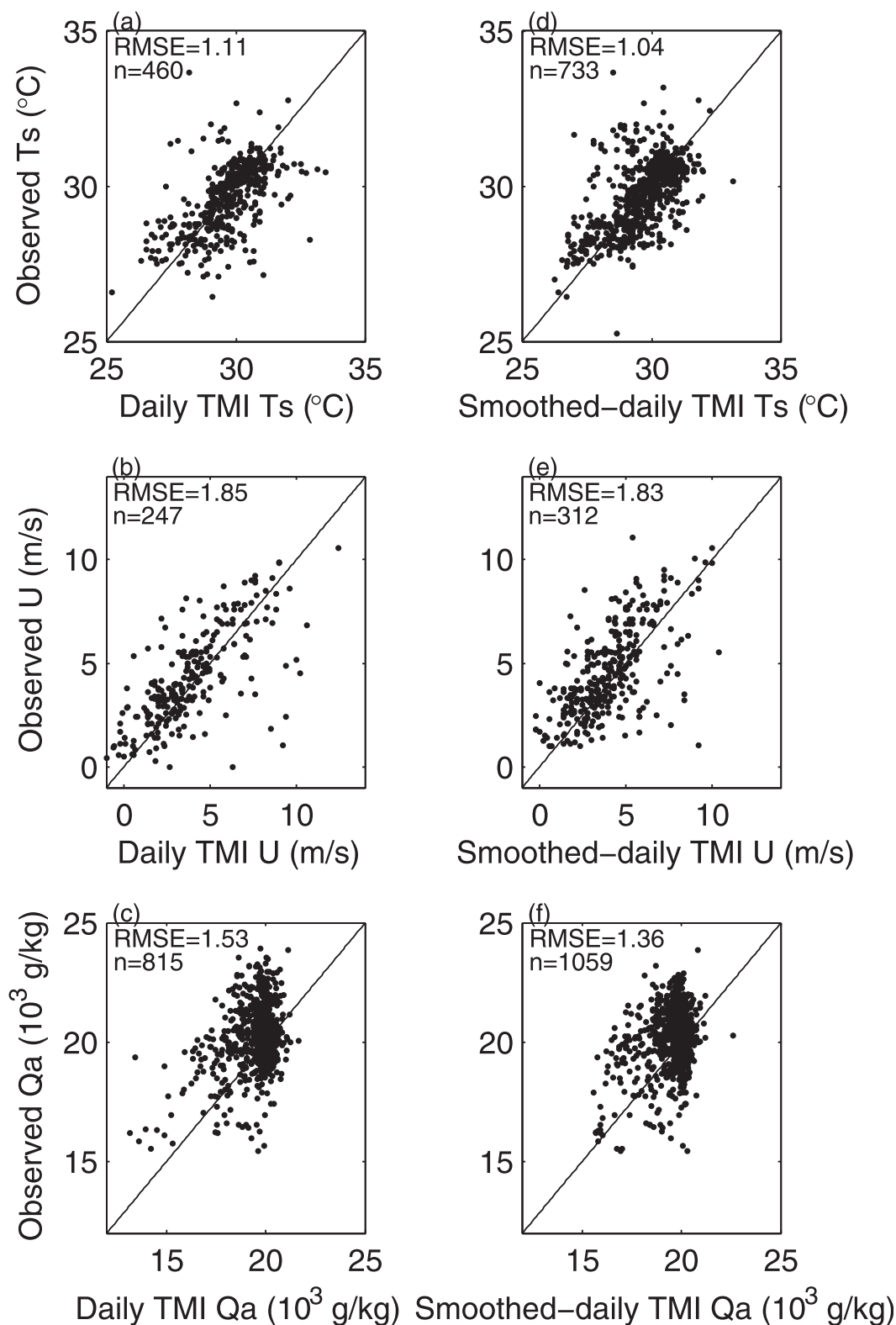


FIG. 3. Scatter diagram of three input parameters: (a),(d) T_s , (b),(e) U , and (c),(f) q_a . (a)–(c) Parameters from daily TMI with observations, and (d)–(f) smoothed daily TMI with observations. The RMSE and the number of samples (n) are given in the top-left corner of each panel.

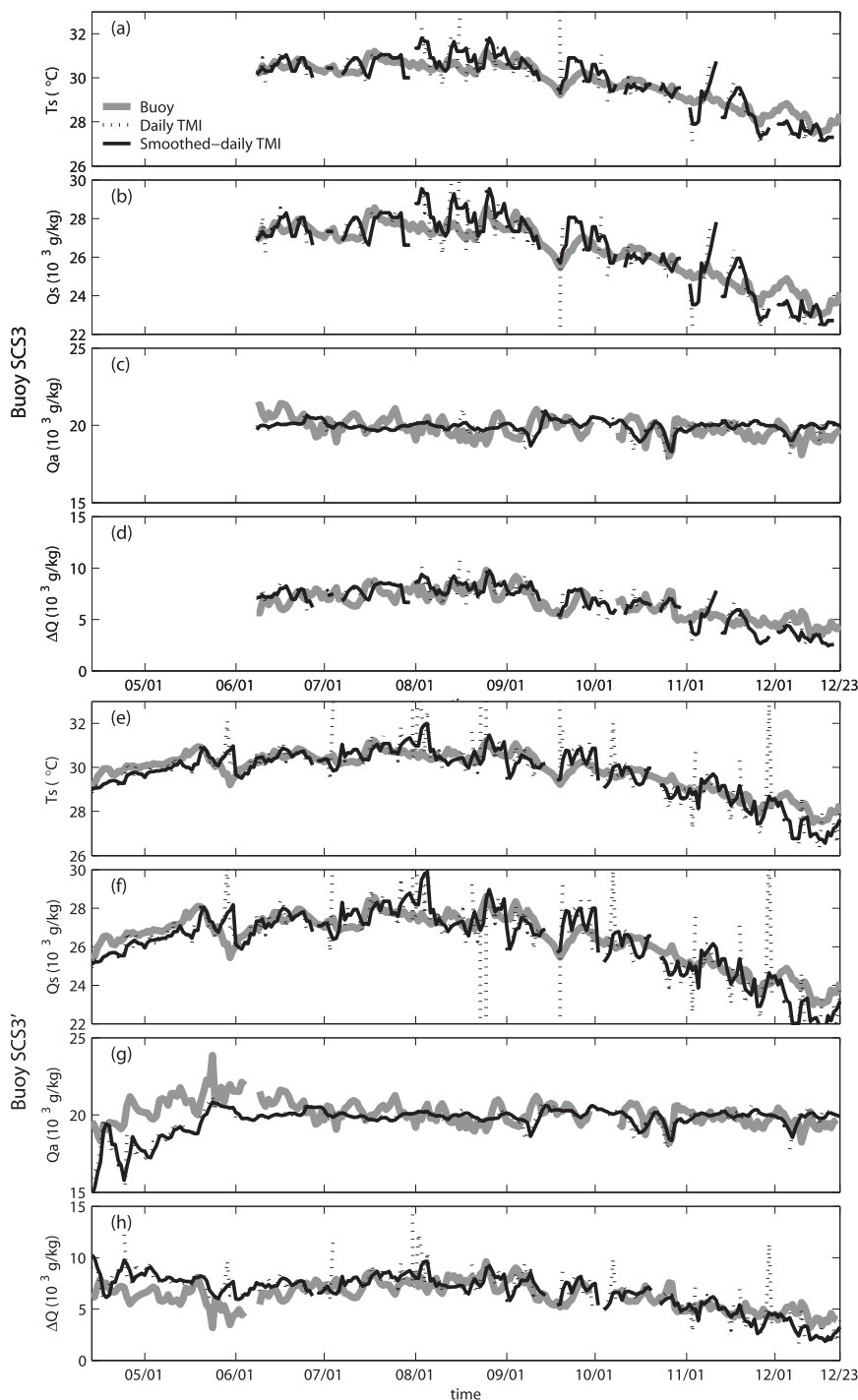


FIG. 4. Daily time series of four input parameters: (a),(e) T_s , (b),(f) q_s , (c),(g) q_a , and (d),(h) $\Delta q = q_s - q_a$ from daily TMI and smoothed daily TMI, and measurements at buoys (a)–(d) SCS3 and (e)–(h) SCS3'.

between observations and (3), the root-mean-square error (RMSE) for each cruise is also computed. If we assume $\rho_a = 1.2 \text{ kg m}^{-3}$, $L_e = 2.46 \times 10^6$, $C_e = 1.2 \times 10^{-3}$, $U = 5 \text{ m s}^{-1}$, and sea-air humidity difference Δq ($\Delta q = q_s - q_a$) = 5 g kg^{-1} in the SCS, then an error of 1 g kg^{-1} in

q_a is equivalent to an error of 17.5 W m^{-2} in latent heat flux according to the bulk parameterization formulas. The minimum RMSE is 1.15 g kg^{-1} , which occurred during the MEL cruise 2006, and gives a bias of 20 W m^{-2} in terms of latent heat flux. Except for the large RMSE

that happened during the Rose cruises, the cruise 2010, and the Xisha cruise, the RMSEs of the other four cruises are all smaller than 2 g kg^{-1} . In summary, the error in latent heat flux using Liu's relationship is mostly less than 40 W m^{-2} when q_a ranges from 12 to 25 g kg^{-1} .

b. Sensitivity to TMI input parameters

As the latent heat flux is predominantly determined by the bulk parameters, it is therefore important to evaluate the sensitivity to these parameters. Figure 3 shows the scatterplots of T_s , U , and q_a , from the daily TMI and smoothed daily TMI; and all the surface meteorological and sounding observations described in Table 1, consisting of 1727 samples. Compared with variables based on the smoothed daily TMI, there are many more missing data based on the daily TMI (Fig. 2). Because the observed winds were often missing at the three buoys, available samples of U are much smaller than those of T_s and q_a . The RMSEs between daily TMI T_s , U , q_a and measurements are 1.11, 1.85, and 1.53, respectively. On the other hand, the RMSEs between these three variables derived from the smoothed daily TMI and measurements are 1.04, 1.83, and 1.36, respectively. This suggests a higher accuracy of smoothed daily TMI data through comparison with a large number of observation samples.

Daily time series of bulk variables from daily TMI and smoothed daily TMI during the deployment periods of buoy SCS3 (12.98°N , 114.41°E) and SCS3' (13°N , 114°E) were also compared with the measurements (Fig. 4). Their development periods, 8 June–23 December and 13 April–23 December 1998, were intended to capture the onset of the winter monsoon in October and the onset of the summer monsoon in late May in the SCS (Li and Wu 1999). Because wind speed was often missing, the evaluation of U was not performed. The four parameters given here are T_s , q_s , q_a , and Δq . There are a lot of extreme values of T_s , q_s , and Δq in the daily TMI data, which could result in large biases. However, these biases are greatly reduced when the smoothed daily TMI data are used in the calculation (Table 2). For example, the RMSE between T_s and ship observations is reduced from 0.66° to 0.56°C , and the RMSE for q_s is reduced from 1.03 to 0.85 g kg^{-1} for buoy SCS3. The reduced biases are similar for buoy SCS3'. The RMSE for Δq at buoy SCS3' is reduced from 1.68 to 1.36 g kg^{-1} . Based on these comparisons, the flux-related variables of the daily TMI data smoothed with a 3-day running mean show a better fit with high-quality observations. This smoothed daily TMI and Liu's q - W relationship are chosen to compute daily satellite-derived latent heat flux over the SCS (called daily SCSSLH). Following the definition for the TMI 3-day running mean, the time of daily

TABLE 2. Comparison of SST, sea surface humidity, surface air humidity, and sea-air humidity difference of daily TMI and smoothed daily TMI from two buoys. Units: for T_s ($^\circ\text{C}$); for q_s , q_a , and Δq (g kg^{-1}).

Buoy TMI data	Corr coef R				RMSE			
	T_s	q_s	q_a	Δq	T_s	q_s	q_a	Δq
SCS3 Daily	0.85	0.84	0.17	0.75	0.66	1.03	0.75	1.23
Smoothed daily	0.90	0.89	0.12	0.81	0.56	0.85	0.69	1.10
SCS3' Daily	0.75	0.73	0.23	0.60	0.83	1.29	1.23	1.68
Smoothed daily	0.89	0.88	0.18	0.68	0.54	0.82	1.21	1.36

SCSSLH is defined on the ending date that contributes to the average.

The ship observations used above were mainly conducted during the summer and autumn seasons. The latent heat flux in the SCS, however, shows a large variability in terms of spatial structure and seasonal cycle. The latent heat flux in winter is considerably larger compared with that in summer. Thus, comparison without the winter season is incomplete at least, and could not provide a convincing conclusion. Since 2008, a new opportunity came with the new Xisha automatic weather station in the SCS. With the help of long time series of meteorological observations, the evaluation of air humidity in winter becomes possible. The comparison of air humidity from daily SCSSLH and OAFlux with buoy/station measurements at buoy QF301 and the Xisha station is presented in Fig. 5. In the northern SCS, the cold-air outbreak situation is indicated by high wind speed, low air temperature, and low air humidity due to the prevailing winter monsoon. This can be readily identified from the time series plots. The timing of the Xisha humidity observations from April 2008 to August 2010 fell within the winter season, and was rapidly intensified in summer. The air humidity from the SCSSLH is more accurate compared to that from OAFlux. During winters of 2009 and 2010, q_a of SCSSLH is also closer to the Xisha observation. However, q_a of SCSSLH is not always more accurate than other daily latent heat flux products. The most obvious deviation happens along the northern continent shelf. At this along-shelf station buoy QF301, q_a of SCSSLH is larger than the measurement from March to May 2011, while q_a from OAFlux shows a better accuracy. Table 3 gives the correlation coefficients R and RMSEs between q_a from SCSSLH and OAFlux daily products and the observations. It shows that q_a from SCSSLH has a higher R and a smaller RMSE than that from OAFlux at the Xisha station. But at buoy QF301, q_a from SCSSLH shows a larger bias than OAFlux. It could be inferred that unlike its high accuracy in the deep areas, the air humidity bias of 2.79 will be attributable to the underestimation of

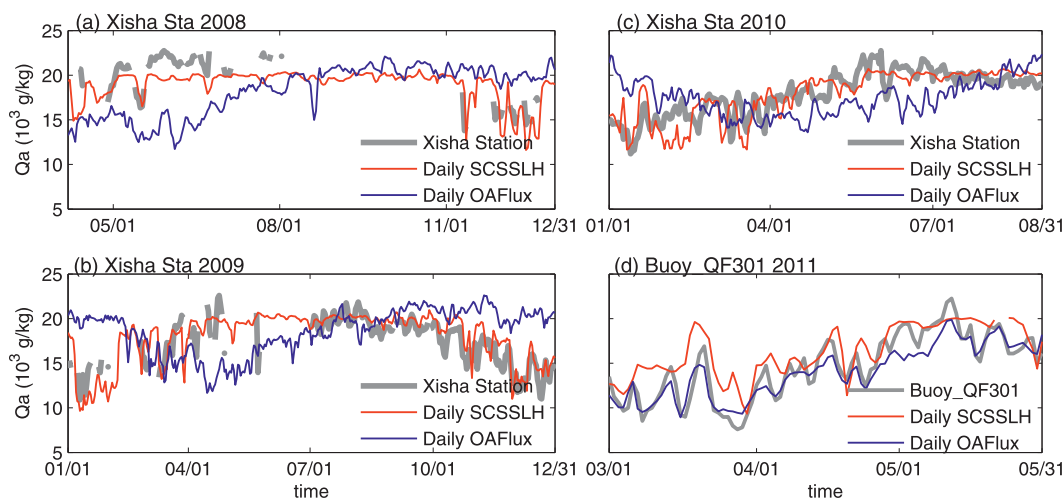


FIG. 5. Time series plots of q_a from daily SCSSLH, OAFlex, and measurements at the Xisha automatic weather station in (a) 2008, (b) 2009, (c) 2010, and at (d) buoy QF301 in 2011.

daily SCSSLH along the northern continent shelf of the SCS.

4. Difference between derived and observed latent heat fluxes

Here, we compare the time series of daily SCSSLH with the ones obtained from two mooring stations and three buoys. Only the periods that contain observed wind speeds were selected for buoys SCS1, SCS3, and SCS3' to avoid dealing with missing wind data. The observed latent heat flux is calculated using the COARE 3.0 algorithm, and surface air humidity is calculated using the relative humidity and air temperature, and the Tetens formula (Bolton 1980).

Among the four daily latent heat flux products obtained by other groups, we find that the ECMWF and NCEP-2 reanalysis products are often higher than the observed values. The satellite-derived fluxes—namely, the GSSTF2, OAFlex, HOAPS3, and our daily SCSSLH—show better performances than the two reanalysis products. The GSSTF2 overestimates the latent heat flux sometimes, especially at station KX and buoy SCS1. The OAFlex is generally closer to the observations than the GSSTF2 and HOAPS3. The daily SCSSLH matches the observed flux best, although it underestimates the flux at station SY. According to Fig. 6, these five products all overestimate the flux during summer in the SCS.

Table 4 gives R and RMSEs between the five daily products and the observations from the two mooring stations and three buoys. Daily SCSSLH has the highest R , except for the mooring station SY, whose R is lower than that of the NCEP-2 product. SCSSLH also shows

the highest accuracy or lowest RSME, except at buoy SCS3, whose RMSE is larger than that of the OAFlex. The maximum RMSE of 32.85 between SCSSLH and the observations occurs at mooring station SY, but it is still the smallest one when compared with other products at the same location.

We also compared the spatial distributions of the latent heat flux products over the SCS. Figure 7 shows SCSSLH and other five products in January and July of 2000, which is a normal year for the SCS (Wang et al. 2006). The large-scale patterns of these monthly-mean fields are similar, but quantitative differences are significant. In January, the two reanalysis datasets (ECMWF and NCEP-2) have high latent heat flux values, while the magnitudes of three satellite-derived products are relatively lower. The SCSSLH shows the minimum latent heat flux in the southwestern basin. According to the preceding discussion in Fig. 5d, this underestimation of daily SCSSLH along the northern continent shelf is mainly

TABLE 3. Comparison of q_a (g kg^{-1}) from SCSSLH and OAFlex daily latent heat flux with the observations from the Xisha automatic weather station and buoy QF301.

	Station	SCSSLH	OAFlex
Corr coef R	Xisha station 2008	0.77	-0.56
	Xisha station 2009	0.82	-0.13
	Xisha station 2010	0.77	0.04
	Buoy QF301 2011	0.81	0.94
RMSE	Xisha station 2008	2.24	5.70
	Xisha station 2009	1.74	4.21
	Xisha station 2010	1.78	3.20
	Buoy QF301 2011	2.79	1.46

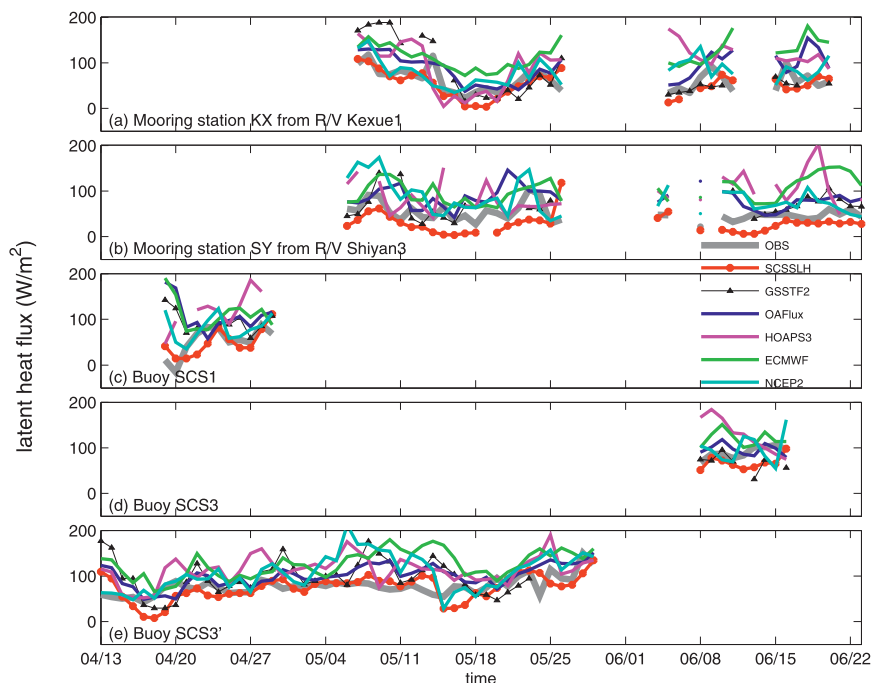


FIG. 6. Time series of daily latent heat flux from observations, SCSSLH, GSSTF2, OAFlux, HOAPS3, ECMWF, and NCEP-2. Stations are (a) mooring station from R/V *Kexue1*, (b) mooring station from R/V *Shiyang3*, (c) buoy SCS1, (d) buoy SCS3, and (e) buoy SCS3'.

contributed by the bias of air humidity. In July, ECMWF and NCEP-2 have rather large values over the basin. Though SCSSLH, GSSTF2, OAFlux, and HOAPS3 are all satellite-derived, the OAFlux is too smooth over the SCS basin. Because of the lack of accurate basinwide verification data, especially in winter, we could not quantify the degree of improvement of the SCSSLH in terms of basin mean.

5. Conclusions and discussion

The latent heat flux is closely correlated to wind speed and humidity difference at the sea surface, both of which

have strong variations on small temporal and spatial scales. Only satellite-based observations can provide a comprehensive estimate of these input parameters. The accuracy of satellite-derived latent heat flux is strongly dependent on the accuracy of input parameters and the parameterization of q_a .

We have acquired a unique and high-quality set of measurements in the SCS. The dataset provides the best opportunity to evaluate the regional applicability of the global dataset and global retrieval methods for latent heat flux in the SCS. Flux-related surface meteorological variables (T_s , U , and q_a) from daily TMI and smoothed daily TMI fit reasonably well with 1727 high-quality

TABLE 4. Comparison of six daily latent heat products with the observations from mooring stations and buoys (W m^{-2}).

	Station	SCSSLH	GSSTF2	OAFlux	HOAPS3	ECMWF	NCEP-2
Corr coef R	Mooring KX	0.60	0.62	0.51	0.35	0.37	0.48
	Mooring SY	0.27	0.06	0.05	0.01	0.15	0.54
	Buoy SCS1	0.58	-0.56	-0.75	0.56	-0.67	0.35
	Buoy SCS3	0.35	-0.47	0.02	-0.73	0.11	-0.03
	Buoy SCS3'	0.74	0.33	0.55	0.51	0.30	0.49
RMSE	Mooring KX	25.11	47.64	40.77	58.07	63.48	33.20
	Mooring SY	32.85	36.04	43.11	59.86	59.00	43.11
	Buoy SCS1	27.40	65.82	82.35	75.21	83.56	43.74
	Buoy SCS3	26.82	32.96	18.01	59.52	35.52	34.67
	Buoy SCS3'	24.97	46.27	32.94	43.85	57.18	43.13

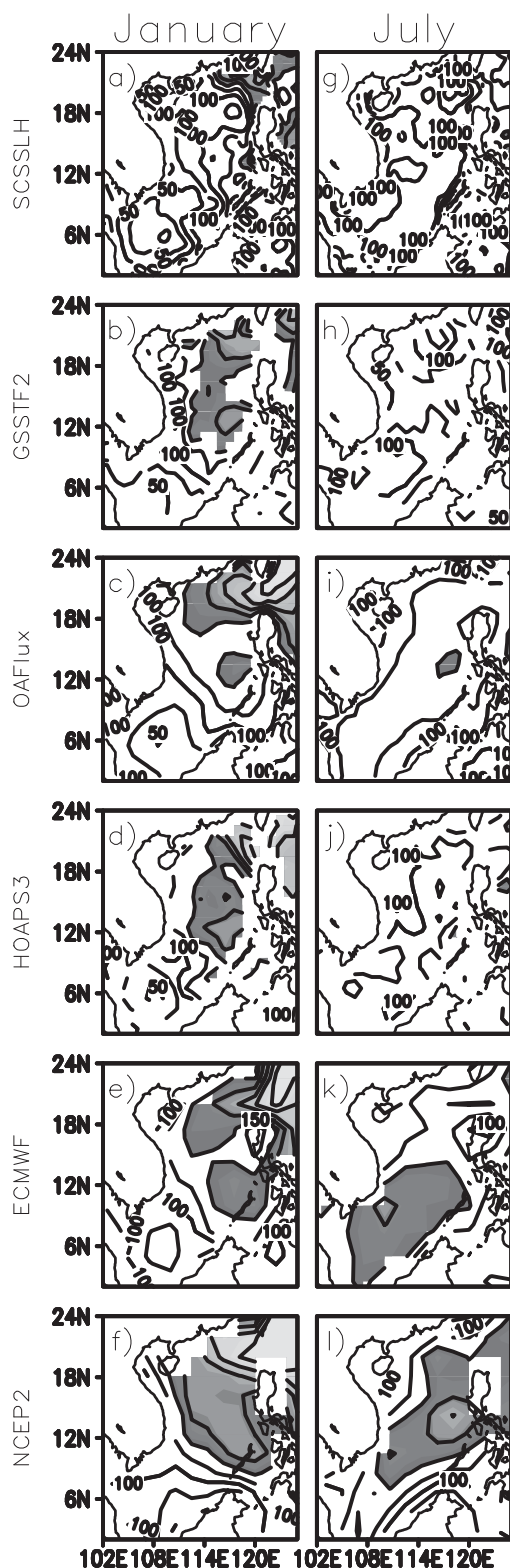


FIG. 7. Daily latent heat flux in the SCS in (left) January and (right) July 2000: (a),(g) SCSSLH; (b),(h) GSSTF2; (c),(i) OAFflux; (d),(j) HOAPS3; (e),(k) ECMWF; and (f),(l) NCEP-2.

observations from seven moored stations and 24 surveys. The smoothed daily TMI input variables agree much better with observations over the SCS basin than those unsmoothed daily TMI data. To ensure high quality of the global fifth-order polynomial near-surface air specific humidity parameterization, verification using 1016 high-resolution radiosonde profiles from 1998 to 2012 is further improved with more sounding observations from the 20 cruises over the SCS from 1998 to 2012. The biases that resulted from the q - W relationship are found to be mostly smaller than 40 W m^{-2} when q_a is in the range between 12 and 25 g kg^{-1} . More importantly, the algorithm validation using the Xisha station observations from 2008 to 2010 and some winter cruises allows us to evaluate the flux during wintertime. These data are used to drive the latent heat flux based on the algorithm of COARE 3.0 with appropriate roughness lengths over the SCS, namely, the SCSSLH product.

The daily SCSSLH, along with the other satellite-derived latent heat flux products of GSSTF2, OAFflux, HOAPS3, ECMWF reanalysis, and NCEP-2 reanalysis, were also compared with high-quality moored buoys and station observations. The daily SCSSLH has the highest special resolution and more realistic values than those from the other five products. This is likely due to the use of improved input variable estimates from TMI, as well as a better algorithm. But daily SCSSLH shows underestimation along the northern continental shelf because of the overestimation of air humidity. Note that the other five products seem to overestimate the latent heat flux systematically in the SCS.

Although our validation analysis based on in situ measurements acquired from buoys and mooring stations has shown that variables from smoothed daily TMI are in better agreement with the in situ measurements, further improvement in the accuracy of latent heat flux will still depend on the availability of high-quality marine meteorological observations, especially in winter. Better meteorological observations and much more high-resolution radiosonde measurements covering wider areas of the SCS (and covering different climatic regimes in other areas) are also desired to allow a more complete regional validation and subsequently to improve satellite retrieval and parameterization of air humidity.

All the satellite products have been undergoing continuous improvement with newer versions being released. The change will accelerate after the issue of the new version of SSMI data by Remote Sensing Systems. All the current satellite-based latent heat flux products rely on the old version of SSMI data, particularly its wind speed, which were wrongly calibrated and will give erroneous temporal variations. Liu and Xie (2008) also demonstrated a new approach, which is different from their

pioneering effort (Liu and Niiler 1984), by retrieving the latent heat flux directly from the radiances measured by microwave radiometer and humidity sounders. This new product achieves much better closure in the atmospheric and oceanic water and energy balances. Unfortunately, their data, which start with the Advanced Microwave Scanning Radiometer for Earth Observing System (AMSR-E) in 2003, do not cover the period in this study.

Our continuous in situ measurements in the SCS should make a contribution to future validations. With the construct of new atmospheric stations on the islands in the SCS, we will have long time series of meteorological observations to propose new algorithms or refine exiting algorithms of q_a . In the case of wind speed, although its impact on the accuracy of latent heat flux is small as compared with that of q_a , it is still important to improve the accuracy of satellite-derived wind speed for better estimation of latent heat flux in the SCS.

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